



Rohit Verma

Bachelor of Technology
in Computer Science Engineering
Indian Institute Of Technology, Ropar

+91-9671098999
rohitverma98999@gmail.com
2023csb1155@gmail.com
GitHub | Website
LinkedIn

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.39 (Till 4th Sem)	2023-2027
Senior Secondary	Central Board of Secondary Education	88%	2023
Secondary	Central Board of Secondary Education	94%	2021

PROJECTS

- **RISC-V Assembler & Simulator** Mar 2025 - Apr 2025
Computer Architecture **Github**
 - Built an assembler to translate RISC-V assembly code into machine instructions with full ISA and directive support.
 - Developed a 5-stage pipelined CPU simulator with configurable data forwarding and hazard resolution.
 - Implemented branch prediction using BTB and 1-bit predictor to reduce stalls during control flow.
 - **Tech Stack:** C++, RISC-V ISA, pipelining, binary encoding, predictive execution
- **SmartCacheSim: Comparing Self-Adjusting and Traditional Caches** June 2025
Data Structures & Algorithms **Github**
 - Implemented and compared LRU (doubly linked list) and Splay Tree caches by caching Dijkstra's shortest-path results on large graphs with randomized query patterns.
 - Analyzed cache hit rates to study adaptive (Splay Tree) vs fixed (LRU) caching strategies, highlighting relevance to real-time systems like traffic routing.
 - **Tech Stack:** C++, STL, graphs, linked list and tree structures
- **AtmosView: Forecast App with Secure API Integration** Jun 2025
Web Development **Github**
 - Developed a weather app using JavaScript to display current and 5–6 day forecasts via the OpenWeatherMap API.
 - Built an efficient Node.js + Express backend to securely proxy API requests and safeguard the API key from client-side exposure.
 - **Tech Stack:** JavaScript, HTML/CSS, basic Node.js & Express, OpenWeatherMap API
- **searchGames: React Vite Game Explorer** Jun 2025
Web Development **Github**
 - Built a React + Vite SPA to fetch and display game data via RAWG API with debounced search and filtering.
 - Developed reusable GameCard components with loading states, error handling, and a secure Node.js + Express proxy to protect the API key.
 - **Tech Stack:** React, Vite, JavaScript, Tailwind CSS, Node.js, Express, RAWG API
- **C-to-Promela: Translator via Gemini LLM** May 2025
Programming Paradigms and Pragmatics **Github**
 - Converts core C constructs into Promela models using Google Gemini LLM and embedded grammar rules.
 - Generates JSON mappings of Promela syntax with explanations for automated SPIN verification.
 - **Tech Stack:** Python, Google Gemini API, regex, JSON

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, JavaScript
- **Development Tools & Frameworks:** HTML, CSS, React, Node.js, Express.js
- **Misc:** Git, GitHub, Vim, MySQL, Risc V
- **Key Concepts:** Data Structures and Algorithms, OOPs, RiscV (ISA)

KEY COURSES TAKEN

- **CSE & Maths:** Data Structure & Algorithm, Computer Architecture, Programming Paradigms and Pragmatics, Digital Logic Design, Discrete Math, Probability and Statistics, Differential Equations, Calculus
- **Others:** Professional Communication, History of Technology, Signal and System, Economics, Biology for engineers, Basic electronics, Tinkering

ACHIEVEMENTS

- **Codeforces (2025):** Achieved a peak rating of 1210 and ranked in the top 20% in three Codeforces contests; view profile.
- **Problem Solving (June 2025):** Practiced and solved 500+ problems across Codeforces, LeetCode, and GFG, strengthening core DSA concepts.
- Won the Esports tournament organized by Infinix at Advitiya Tech Fest 2025.